

- 1 (a) The aye-aye, *Daubentonia madagascariensis*, is a primate native to Madagascar. Aye-ayes are nocturnal (active at night) and make their nests high up in trees. They feed on insect larvae in the trunks of trees.

Fig. 1.1 shows an aye-aye.



Fig. 1.1

The International Union for Conservation of Nature (IUCN) is the world's largest global environmental organisation. The IUCN Red List of Threatened Species™ evaluates the conservation status of plant and animal species.

The aye-aye is categorised as endangered on the IUCN Red List, which means that it faces a very high risk of becoming extinct in the wild.

- (i) Name the domain to which the aye-aye belongs.

.....[1]

- (ii) Suggest **one** reason why aye-ayes have become endangered.

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- (iii) Suggest ways in which zoos may help to protect this species from extinction.

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.....[3]

Question 1 continues on page 4

- Fig. 1.2 is a map of Madagascar showing the location of the two main populations.



(i) With reference to Fig. 1.2, suggest why there is a large genetic difference between the two populations.

[4]

(ii) Name the type of speciation that may be occurring.

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.....[1]

(iii) Suggest **and** explain a pre-zygotic isolating mechanism that could prevent successful reproduction between aye-eyes of the two populations.

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[Total: 12]